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A

For the net force to be acting to the left, the charge Q needs to exert a force on X to counter the downward force on X by Y.

Thus Q is negative.

The vertical component of force on X by Q = force on X by Y

$$k \frac{Qq}{(\sqrt{2}r)^2} \cos(45^\circ) = k \frac{qq}{r^2}$$

$$Q = 2.8q$$

Thus $Q = -2.8q$

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D

Work done on the charge by external agent = change in potential energy